# A Survey of Machine Learning Based Anti-Money Laundering Solutions

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**Due to the proliferation of financial technology and new methods of transaction, we are witnessing the exponential growth of financial transactions across the globe. While current financial infrastructures have been designed to handle large volumes of transactions and to serve millions of clients every day, they fail to accurately identify obscured fraudulent financial activities such as money laundering. Money laundering refers to the act of disguising the source of illegally obtained funds. Among different novel and sophisticated technologies to tackle money laundering, machine learning offers promising results in discriminating suspicious transactions from normal activities. This survey provides a comprehensive overview of recently published works about anti-money laundering solutions that rely on machine learning. It further provides a detailed summary of their underlying machine-learning techniques and algorithms. Since data is the most crucial part of every Anti-Money Laundering solution, this survey provides a comparison among the types of data used in the models we studied. Additionally, in this survey, we introduce a new taxonomy based on the type of data used to train and test each model. These data types cover transactions, customer profiles, customer relationships, and transactional behaviors. Lastly, we provide an analysis of the performance and limitations found in existing Anti-Money Laundering works and compare the effectiveness of different machine learning techniques with the objective of finding potential research gaps in this area.**

***Index Terms*—Anti-Money Laundering, Customer Profiles, Customer Relationships, Data Mining, Machine Learning, Transactional Behavior**

## I. INTRODUCTION

THE rapid growth of modern technology, digitization, and global communication is a two-edged sword. Although they drastically improve our lives in numerous aspects, they also make it easier to commit fraud at a much greater rate than ever before, inflicting severe losses on individuals and organizations [1]. Most of those who involve in illegal activities have a single main incentive: money [2].

Money plays a vital role in devising and accomplishing fraudulent operations. For example, the annual income of criminal organizations from human trafficking is about $150.2 billion USD, which is obtained through actions such as sexual exploitation, forced labor, and organ harvesting [3]. In human trafficking, money is the first and most essential requirement to move victims between different locations, handle the places and procedures related to those victims beings exploited, and bribe those who act as interceders condoning this criminal action. To strive against the motivators of criminals and their ability to participate in unlawful activities, the strong relationship between money and crime should not be underestimated. Therefore, governments should impose tough restrictions on money flowing through criminal institutions [4].

According to the United Nations Office on Drugs and Crime, money laundering is the process of dissembling the origin of money obtained from illicit tasks without compromising launderers who are the primary beneficiaries [5]. The approximate amount of money laundered globally through a year is 2%-5% of global gross domestic product (GDP) [6].

Criminals take advantage of the vulnerabilities found in financial institutions to hide the patterns of illegal transactions. Hereby, law-enforcement officials can hardly identify suspicious cases as they appear as benign transactions. Financial institutions are generally at the center of money laundering activities. They must possess the necessary infrastructure and tools to serve high-risk clients while avoiding fraud and money laundering activities.

Anti-Money Laundering (AML) is a known name assigned to these programs. In fact, AML refers to the laws, regulations, and procedures intended to prevent criminals from disguising illegally obtained funds as legitimate income. It seeks to deter criminals by making it harder for them to hide the source of dirty money [7].

Banks are suffering from a lack of effective AML solutions to combat threats stemming from money laundering activities [8]. In the manual detection of suspicious activity, an employee of the bank must promptly create a report describing the suspicious activities when observing an unusual behavior of customers [2].

The push to utilize online services has paved the way for accomplishing a higher number of financial transactions without the need for in-person interaction. Consequently, the rapid growth of companies providing financial-related technologies has indicated that the trend of digitization has also been amplified over the recent years. This ever-increasing volume of financial transactions necessitates an effective monitoring system in order to put transactions under surveillance in real-time. This in turn may highlight the importance of providing updated and automatic AML systems for banks to combat suspicious cases. Accordingly, the automatic transaction monitoring system should always be taken into account as an obligatory part of any AML scheme [9][10].

Traditional transaction screening systems work based on a hierarchical process to identify money laundering cases through various layers[11]. In the first layer of the hierarchy, suspicious transactions are flagged based on multiple predefined rules as well as clustering analysis. Then, a risk score is assigned to each flagged transaction. In the second layer,

those flagged transactions that have a risk score above a threshold are examined by an experienced analyst. The transactions that are selected by the analyst are chosen to generate Suspicious Activity Reports (SAR), which are then submitted to the Financial Crime Enforcement Network (FinCEN) [12].

In the past few decades, conventional transaction monitoring systems have been based on embedded rules, a simple set of if-then conditions with predefined thresholds using historical data or business intuition developed by consultants and domain experts [13]. However, one of the main drawbacks of a rule-based system is its reliance on a set of rules which should be continuously validated and fine-tuned. This is problematic as these rule updates require a time-consuming and labor-intensive process. The process of updating these rules is imperative because money launderers constantly devise new strategies to avert the detection rules. Moreover, as we are dealing with dynamically changing data, integrating new and existing rules may undermine the system's performance [14]. As well, these models are not robust enough to detect challenging and complex transaction patterns because they merely focus on single transactions and simple patterns.

Rule-based systems work perfectly with structured and constrained data. Still, it would be significantly challenging for them to deal with the bulky dataset containing various kinds of structured, semi-structured, and unstructured data [15]. Conversely, machine learning models show outstanding performance when they are exposed to new unseen data, given that they are properly trained. The trustworthiness of their results and decisions are rooted in their ability to learn from previous computations. To put it another way, machine learning algorithms can effectively interpolate and extrapolate on new scenarios without being taught every possibility by recognizing and applying patterns learned in the training stage [16].

Moreover, rule-based methods produce no result if data contains complex money laundering patterns. At the same time, machine learning models are designed to capture nonlinear relationships between dependent and explanatory variables. Compared to machine learning methods that can automatically learn the decision boundaries, an expert-knowledge is an essential part of each rule-based framework to create new rules [17]. Having said that, the ability to manage and analyze sizeable unstructured information, self-update with anonymous data, and recognize complicated patterns make machine learning an effective and efficient choice for detecting money laundering.

As a result of the proliferation of technology, the volume of financial transactions has grown significantly, and the methods by which transactions are conducted have become more complex. With this in mind, rule-based methods are becoming more and more ineffective as they are not able to automatically adapt to these changes.

Recent research on AML shows that a better approach to identifying money laundering activities in financial data is by applying machine learning methods. While there are a few survey papers on Machine-Learning-based AML (ML-based AML) solutions [1][18][19][20], they only classify models based on the general groups of machine learning (supervised, unsupervised, and semi-supervised). Since the type of input data plays a vital role in machine learning techniques, our survey provides a different perspective and taxonomy based on data type compared to existing survey papers on the same topic. This will help researchers to have a better insight into how these models behave with different data types and which types of data have been mostly used among various works. We cover the majority of works published in the area of AML during the recent decades.

The remainder of this paper is organized as follows. Section II describes the definition of AML and related challenges. An overview of data preparation and machine learning techniques primarily used in the field of money laundering detection is provided in Section III. Then, we introduce our new taxonomy of AML approaches in Section IV and explain the literature review. Section V states the analysis of the literature work. To better understand the existing solutions, we compare the current commercial methods of AML in Section VI. Finally, the conclusion of the survey and some of the future research directions are summarized in Section VII.

## II. Definition of Anti-Money Laundering

In this section, we explain the concept of AML in-depth and then provide an overview of the challenges we face in this field. Money laundering consists of three main phases namely: *placement, layering,* and *integration* [21].

Placement is the action of converting fraudulently made funds into a financial system such as a bank. Layering is the process of transferring money to other banks or individuals through money orders, cheques, wire transfers, or other methods to conceal the origin of the funds. In the final step, funds are placed into the market by purchasing legitimate instruments. Therefore, this step is known as integration. This is where the dirty money is merged with clean money and entered into the economy. Depending on the available data, AML techniques may be applied to any of the above three phases.

In order to reduce fraud and combat criminals, as well as to conform to various regulations, the financial institutes must diligently identify, track and report fraudulent financial activities. These institutes handle a large number of transactions on a daily basis. The large volume of data and complexity of transactions makes it challenging to accurately and efficiently identify money laundering activities.

With this goal, AML is introduced as a solution for financial institutions to fight against money laundering. AML provides methods to analyze customer profiles and activities to extract valuable information. This information includes abnormal activities such as significant increases or withdrawals performed and is used to discover money laundering cases. Generally, AML involves all the laws, policies, procedures, and regulations that put pressure on banks to keep an eye on their clients and try hard to prevent money laundering. In the following paragraphs, we will briefly discuss each of the challenges in AML.

Financial services must comply with various regulatory frameworks and privacy policies, thus access to extensive

financial datasets is not trivial. Due to this, the U.S. documentation of machine learning applications in AML is incomplete, resulting in insignificant and insufficient information about data, machine learning models, and their performance [22].

Dealing with rare events is another fundamental challenge that should be kept in mind while modeling money laundering detection frameworks. The situation where the number of money laundering events is significantly fewer than the non-money laundering events implies the rarity of money laundering events. Therefore, in this survey, we refer to money laundering events simply as events and non-money laundering events as non-events.

False-negatives (inaccurately classifying an event as non-event) and false-positives (inaccurately classifying a non-event as an event) are two major challenges resulting from the aforementioned rare event issue [23]. In this scenario, false-negative errors carry a much higher cost than false-positive errors. Therefore, the ability for the model to make fine distinctions will likely be inadequate if we neglect the cost differences associated with different classification errors. In turn, this negatively affects the model's ability to efficiently and precisely identify various patterns in the data and classify those events into their correct groups. Consequently, the model will perform sub-optimally when event frequency is low.

Generally, there are two groups of methods to solve the problem of rare events, both of which are model-independent [24]. The first group revolves around adjusting the algorithm, such as modifying the misclassification costs for different classes. The second group consists of methods that manipulate data before using it in the model. This pre-processing approach is comprised of two main techniques and aims to augment the events with additional information to create a more balanced dataset.

The first technique is to change the target variable. In this method, an alternative target variable is chosen, closely related to the original target variable, and has a higher frequency [24]. Unfortunately, the alternative variable is not always on hand and may not reproduce the exact relationship. To address this issue, the second technique known as Sampling is suggested.

This approach is frequently used to address the rare event data issue. At the high-level, sampling refers to the methods of increasing the rare event rates through two main approaches: under-sampling the majority group and over-sampling the minority group [25]. The following section will briefly investigate various techniques for making data more handy and effectual and summarize machine learning models mostly used in the AML field.

## III. Overview of Data Preparation and Machine Learning Models

The definition of a successful detection framework refers to a system capable of discovering trends or patterns of suspicious behavior by generating a signal which shows the potential of a money laundering action. This generated signal has a score that determines whether the case should be investigated further[26][27]. The process of suspicious behavior identification begins prior to opening an account possibly via a background check of a customer. Once the account is opened, it starts to monitor the behavior of the new account.

In order to flag suspicious transactions, a successful AML solution should give a detailed and precise reasoning. As a result of the efficient identification, monitoring, and management of the risks associated with money laundering activities, a comprehensive report will be generated. To achieve these goals, the AML solutions used to identify suspicious transactions should have the following four imperative features: data quality, scalability, detection accuracy, and reaction time.

The term data quality refers to how well a dataset fits its specific context. In order to assess the data quality, we need to measure data quality characteristics such as accuracy, completeness, consistency, timeliness, validity, and uniqueness. However, the first four characteristics have been widely cited in the literature as the most critical data quality attributes [28].

There is a wide range of definitions for accuracy, but it is mainly considered as meaning a correct and unambiguous correspondence with the real world. Completeness indicates the ability of the dataset to deliver all the required values that are available. The uniformity of data as it flows between networks and applications is referred to as data consistency. The identical data values stored in different locations should not be in conflict. Finally, timeliness can be defined as the degree to which data is outdated.

Scalability determines the lifetime value of an AML model. A system is said to be scalable if it doesn't require to be rebuilt in order to retain effective performance during or after a significant increase in workload. By workload, we mean the number of concurrent users, physical storage capacity, the maximum number of transactions handled, or anything else that exceeds the system's initial capability. For example, it is common for a large bank to deal with several millions of transactions on a daily basis. As a result, scalability is critical to ensure that the system continues to function efficiently as the volume of data grows [29].

One of the essential features of any detection system is its detection accuracy. It refers to the ability of a system to diminish the number of false positives while keeping the detection rate at a reasonably high level. There is a strong correlation between data quality assurance and detection accuracy because it is not possible to obtain acceptable accuracy without a great amount of data quality assurance[30].

A practical AML solution should be able to process tens of millions of transactions in a matter of hours, minutes, or seconds, this is called reaction time. In the following subsection, we will discuss the data preparation phase of AML. We then describe the common machine learning approaches in the context of AML.

### A. Data Preparation

Data is the key for banks and financial institutions to develop an effective model for money laundering detection. The process of gathering, combining, structuring, and organizing data to make it usable by business models is known as data preparation. Although data preparation is a time-consuming process for data professionals, it is necessary as a prerequisite

in order to put data into context and reduce bias caused by poor data quality.

One of the primary purposes of data preparation is to guarantee that raw data being prepared for data processing and analysis is accurate and consistent. As a result, models which have been designed to work with this data will produce valid outputs. However, data is commonly created with missing values, inaccuracies, or other errors. In addition, datasets can be obtained from different sources with different formats that need to be unified. Correcting data errors, verifying data quality, and joining data sets constitutes a big part of the data preparation process. While data preparation includes data pre-processing, profiling, cleansing, and transformation; often it also involves pulling together data from different internal systems and external sources. In the following, we will briefly describe each component of data preparation.

Generally, relevant data is collected in raw format from many sources, making analysis impossible. Data pre-processing is a step in most machine learning processes where the plain structure of data gets transformed and encoded to be simply parsed by the machine. The majority of machine learning models we have reviewed perform better with organized datasets than raw data. That is to say, pre-processing raw transaction data will help the model better recognize the transaction behaviors of bank clients and identify suspicious patterns of financial transactions.

Data profiling refers to the process of evaluating and analyzing data to understand how it's structured, what it contains, the relationships between datasets, and how it could be used most effectively. It aims to identify patterns, anomalies, inconsistencies, and other issues in the dataset. In addition, data profiling provides critical insights on the dataset that financial institutions might leverage to their benefit.

Data cleansing plays an important role in the overall process of data preparation. It corrects the identified errors in the dataset to create a complete and accurate dataset that is ready to be processed and analyzed. There are three types of errors; faulty data, inconsistent values, and missing values. Faulty data such as strange naming conventions, typos, or incorrect capitalization can either be removed or fixed. Inconsistent values should be harmonized by standardization of the values.

The problem of null values instead of valid feature values is referred to as missing values. This can happen for various reasons, such as irrelevance, omission, or inapplicability of data within a particular context. In addition, some customers are reluctant to share their private information such as annual income amount or their residential address to the financial institute. The performance of certain machine learning models that are susceptible to missing values is significantly affected when there are several missing values in the dataset. There are two general approaches to addressing the missing values issue. Either by getting rid of all transactions with missing values or populating all missing values using algorithms such as mean substitution, cold deck imputation, and hot-deck imputation [31].

Data transformation is another component of data preparation where data structure, value, or format may be affected. In other words, transformation can occur by creating structured or semi-structured data from a non-structured dataset, normalizing the values of data within a specific range, and changing the format of data to be unified. In fact, data transformation is a key step in preparing data as it provides a consistent dataset usable by machine learning models.

Generally, data preparation is a tedious and time-consuming part of designing machine learning models [32] for any AML process since AML models should fit the large dataset of transactions. It is a challenging step, yet only a small number of existing publications have proposed methods to address it [33][34].

Besides data preparation components, we can evaluate financing actions within different time intervals because money launderers may choose certain times during the day, week or month to commit a crime. Also, criminals may frequently change their behavior and methods. Hence, a reliable AML program should be updated consistently to keep pace with changing requirements, we call this adaptability. However, requiring adaptability from a model results in a massive cost for maintenance and repair of the system.

In the following, we will review the common machine learning methods used in the context of AML

### *B. Machine Learning Models*

Machine learning is a branch of artificial intelligence capable of automatically learning from experience without detailed prior knowledge. In other words, it gives computers the ability to self-learn and adapt independently to new data without human intervention [35]. In recent years, machine learning has shown a considerable number of successful applications in different subject areas [36] [37] [38].

Researchers have recently proposed a large number of approaches for AML using machine learning models. Although it is difficult to investigate all published works in this domain, we have tried our best to present the majority of them that apply machine learning as the core part of their overall process of AML. It is essential to analyze, review, and compare various methods of money laundering detection to be able to recognize laundering crimes, patterns, abnormal behaviors, and money laundering groups. Generally, machine learning approaches are divided into three main groups: Supervised, semi-supervised and unsupervised.

Supervised machine learning models learn gradually from a training dataset. They use a labeled training set containing both clean and suspicious transactions and required patterns or features to create the detection model. When the training phase is completed, the model can predict new unseen samples based on the knowledge obtained from past data. In supervised mode, the dataset must be structured dramatically to apply a machine learning algorithm. These techniques seem to be well-suited for banks who have experience dealing with ML-based detection models.

Semi-supervised refers to those algorithms that learn from both labeled and unlabeled datasets. The labeled dataset involves banking records and their associated labels, which help the model to discern patterns and discover the relationships. An unlabeled dataset is used to identify unobserved patterns in

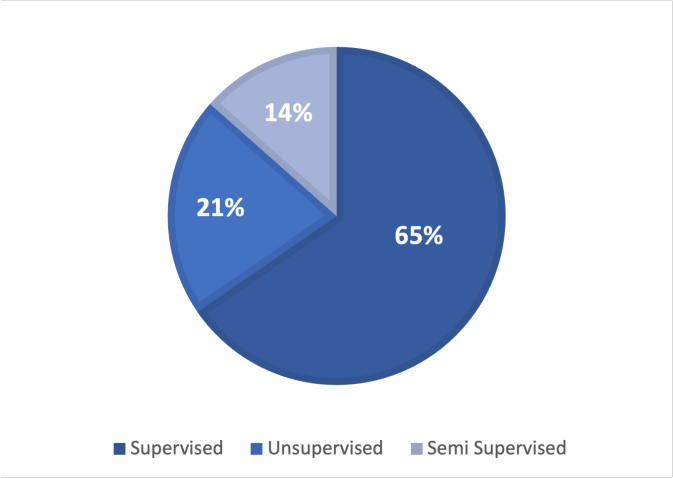



Fig. 1. Distribution of existing works based on the type of machine learning model

addition to the trained dataset. These algorithms consider new transactions as invalid if they match with none of the training set records. Semi-supervised techniques are a proper fit for those banks that already have a system to classify regular and anomalous transactions.

Unsupervised machine learning approaches consume unlabeled datasets. This set of machine learning techniques relies on their model without any prior knowledge of the data or human supervision. The objective of unsupervised learning is to identify hidden structures and patterns as well as similarities among the data records. Since this set of techniques does not require any labels or training data, it could be a good candidate for financial institutions that are not capable of examining and labeling their data. Figure 1 depicts the distribution of machine learning model groups observed in our surveyed publications. The following paragraphs provide a brief explanation of machine learning techniques mostly used in the literature review.

**Decision Tree**: The decision tree is a robust supervised learning algorithm mainly applied for classification and regression purposes. The motive of the decision tree is to forecast the value of a target variable according to the learning decision rules derived from the dataset. Nodes, branches, and leaves are the three fundamental elements of a decision tree structure. A node represents a predicate on a particular feature. A branch represents the possible results of the predicate, and leaf is the label of available classes [39]. Unlike artificial neural networks, which only deliver the results without explanations, decision trees explain their predictions through a series of rules. Moreover, the decision tree can work with non-numeric datasets. A decision tree classifier can precisely fit a set of training samples by constructing a sequence of rules that form a pattern.

Two major benefits of the decision tree and its variations are clarity and interpretability. Unlike other learning methods, the motivation behind selecting each decision can be directly interpreted for human understanding. In the case of AML, a clear explanation behind a decision has many benefits. Primarily it can assist with producing evidence supporting a decision made by the model, as well as ensuring the decisions made adhere to the procedures set by the financial institute.

On the other hand, a decision tree with a complex structure and learning pattern without significant enhancement can easily lead to model overfitting. As a result, the pruning technique should then be applied to simplify the decision tree by omitting the nodes with slight learning improvement. The architecture of a decision tree for massive real-world datasets must be articulated and designed correctly to avoid encountering scalability issues [40].

**Random Forest**: Random forest is an ensemble method that is built based on a multitude of decision trees. This method can be applied to both classification and regression problems. For classification problems, the random forest output is the class chosen by most trees. While for regression problems, the mean or the average of the values obtained from individual trees is considered as the result.

In comparison to a single decision tree, random forests can lessen the risk of over-fitting while increasing the overall accuracy. However, this success is heavily reliant on the usage of uncorrelated decision trees because if a random forest uses identical or extremely similar trees, the outcome will be very close to that of a single decision tree. Bootstrapping and feature randomness [41] are two techniques that are used by random forests in order to achieve uncorrelated trees. Furthermore, decision trees in a random forest model run in parallel, ensuring that the performance cannot be affected by a single tree.

While random forests generally outperform decision trees in terms of accuracy, they do so at the expense of decision trees' inherent interpretability. The main reason behind this lack of interpretability is the layout of random forest, which includes hundreds of deep decision trees, making it as a "black box" [42]. Similar to decision trees, random forests can grow quite complicated when dealing with huge datasets. Pruning the tree can help reduce the complexity, but it can also result in parts of the tree being removed that contain critical information.

**Artificial Neural Network**: An artificial neural network is a set of interconnected nodes that resembles the inner neural structure of the human brain. A neural network is commonly used to solve clustering and classification problems [43]. Their main objective is to uncover obscure features that are not obvious to humans to identify. An artificial neural network is made up of three layers.

The first layer is known as the input layer, and the third layer is called the output layer. The middle layer, also known as the hidden layer, contains the neurons that obtain values from the input layer. Neurons in the hidden layer take a set of weighted inputs and produce the result through the activation function. The result of the middle layer is then passed to the output layer [43]. Generally, the number of neurons in the output layer is equal to the number of different classes for a specific machine learning problem. In the case of AML, the neural network will have two neurons in the output layer identifying clean and suspicious transactions.

Deep learning is a sub-category of neural networks with multiple hidden (deep) layer architectures. Deep learning models require a large volume of training data to complete the

learning process. Therefore, these types of models are suitable for financial use cases where a significant amount of data is at their disposal. Although deep learning models surpass most other machine learning models in terms of accuracy, they are generally referred to as "black boxes" because it is hard to determine exactly what procedures are performed internally to arrive at a certain decision [44]. This could be problematic in AML use cases, where it is required to clearly understand how a financial transaction is flagged as suspicious.

**Support Vector Machine**: Support Vector Machine (SVM) is a flexible group of supervised machine learning techniques that has applications in regression and classification [45]. This group of techniques describes a method to identify a separator super-vector of data points with a maximal margin between the two classes. A margin is defined as the distance between two classes established by the super-vector[46]. An SVM model has the ability to convert a nonlinear issue space into a linear problem in a higher-dimensional space. Because of SVM's capacity to generalize on high-dimensional space problems, it's an intriguing solution for the AML use case.

According to the reviewed publications, standard SVM is severely reliant on the training dataset, showing adequate performance only on small to medium datasets. The performance of an SVM suffers when handling a large volume of data, as it's the case with AML. The accuracy of the SVM technique can be enhanced in several ways, including training on additional data sources, employing rigorous feature selection approaches, substituting the internal kernel with one more suitable, and employing ensemble classifiers.

**One-Class Support Vector Machine**: While classic SVM is primarily used in cases where there are multiple classes, the One-Class SVM (OCSVM) supports a single class. Generally, the OCSVM model is helpful in identifying outliers as it makes it possible to separate representative samples from anomalous data samples. While a useful mode, OCSVM suffers from loss of accuracy when there are a large number of anomalies [47].

**Naïve Bayes**: Naïve Bayes is a collection of classification algorithms that quickly utilizes Bayes conditional probability rules. It is a family of algorithms sharing identical principles, not a single algorithm [48]. Assuming that features are independent, the Naïve Bayes assigns a class to new observations that maximize its probability given the class label. Also, each class label is considered as a random variable in this algorithm. As a result, naïve Bayes models can be built easily without any complicated iterative parameter estimation. This is considered as a key advantage that makes them practical for dealing with large datasets.

**K-Nearest Neighbor**: K-Nearest Neighbor (KNN) is a common type of clustering algorithm that can be used for classification and regression use cases. At the high level, KNN attempts to find the samples that have the highest similarity among their feature set [49]. New unseen samples are assigned to most similar clusters previously formed during the training phase. There are three phases in the K-Nearest Neighbor algorithm: a) calculate distance, b) find the nearest neighbors, and c) vote for class labels. According to recent research results [50], KNN has proven to be an effective algorithm for minimizing the false alarm rates and increasing the fraud detection rate.

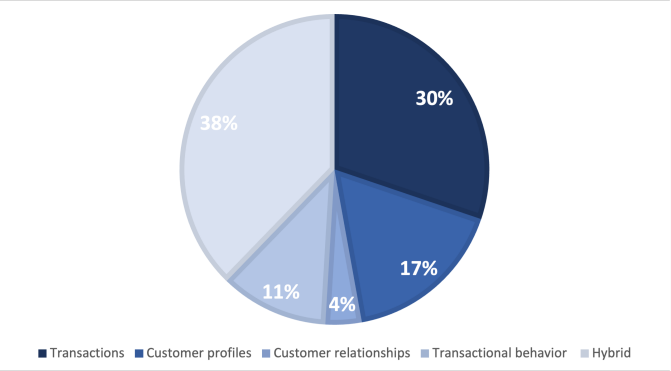



Fig. 2. Distribution of existing works based on the proposed taxonomy.

**K-means Clustering**: K-means is an unsupervised machine learning model capable of clustering similar data instances and identifying hidden patterns [51]. This algorithm's purpose is to detect groups of similar data, group them together, and uncover hidden patterns. To achieve this object, K-means attempts to learn from features of data and search for a pre-defined number ($K$) of clusters in an unlabeled dataset. Then, it defines the value $K$ to indicate the number of centroids. It then produces $K$ number of clusters in the dataset by assigning each data sample to a cluster based on the smallest delta between the cluster's centroid and the data set. From there, it designates and averages data that are close to a given centroid in order to discover the proper cluster for each data.

Anti-money laundering solutions based on AI are a common research topic, particularly in recent years. Financial services process large volumes of transactions on a daily basis. Furthermore, there is a complex web of relationships among the transactions and the accounts. Therefore, a practical AML solution must be able to handle the large and complex amount of financial data efficiently. With many algorithms to choose from, machine learning is a viable solution capable of addressing these problems effectively. The following section states the literature review of AML and introduces the proposed taxonomy of this survey to classify ML-based AML approaches.

## IV. Literature Review

The publications reviewed are categorized based on their approach to machine learning into three groups, namely supervised, unsupervised, and semi-supervised. The publications are further categorized based on their use of real versus synthetic data sets. Moreover, to classify the publications more granularly, we introduce a new taxonomy on the type of input and output data used by proposed models. Our taxonomy includes transactions, customers, customer relationships, and transactional behaviors. The following are the types of data identified in our literature review and the description of each taxonomy. Tables I, II, III provides a comparison of existing works using supervised, unsupervised, and semi-supervised techniques, respectively.

### A. Fraudulent transactions and clean transactions

This group of papers works directly on transaction information. Transactions have a wide range of attributes that each of them provides different data for performing detection systems. The methods of this category take transaction-related data such as the amount received, amount withdrawn, or the frequency of transactions over a certain period of time (daily, weekly, monthly). The output tries to identify whether the desired transaction is suspicious or benign. The following paragraphs describe the approaches of different research works in this group.

To distinguish suspicious transactions from legitimate transactions, Lv et al. [52] designed Radial Basis Function (RBF) network. RBF technique utilizes artificial neural network architecture, putting nodes in different layers together for cognitive learning. Their proposed network contains weights that determine the strength of the connection between two adjacent nodes. The main purpose of their model is to determine the center and width of RBF before calculating the weights. The overall process for learning RBF network parameters is divided into two parts.

First, instead of utilizing typical K-means clustering, they used the APC-III clustering [53] technique to identify the parameters of hidden layers in the RBF network. The centers of RBF neural networks are often determined via K-means clustering, where the center is typically acquired by randomly selecting points from among all of the data points, resulting in a somewhat poor learning speed. APC-III clustering can make full use of the information in training data, and it achieves a faster learning speed by clustering the patterns class by class rather than the entire patterns at once. Because the mapping from hidden layer to output layer is linear, the determination of the weights for the second half of their approach becomes a linear optimization problem. Hence, they applied Recursive Least Square (RLS) algorithm to update the weights instead of gradient descent due to its low speed.

In another article, chen et al. [54] introduced the use of the Expectation Maximization (EM) [55] algorithm in AML. The goal of their research is to take advantage of EM's effectiveness in the detection of suspicious transactions. The data used in this study came from a Malaysian bank. Genetic search and the best first search algorithm were used to pick subsets of critical attributes. For example, the amount, number of credit and debit, and their sum are all crucial fields for the clustering phase, according to their findings.

It is pointed out by chen et al. [54] that real money laundering cases were primarily identified from daily and weekly records based on historical performance. Their experimental results revealed that EM could reduce almost half of the number of false positives while still detecting true positives. In comparison to K-means, the performance of EM was more promising in grouping the data. In fact, EM proved to be significantly effective when dealing with highly unbalanced data. The paper's conclusion was that EM could be a viable approach for clustering in AML with a low false-positive rate.

Alkhalili et al. [56] presented a model that automates the process of verifying blocked transactions in a watchlist filtering system. They developed a machine learning component (ML-Component) consisting of monitoring, advising, and taking action phases. By replacing traditional rule-based systems, their proposed technique aims to reduce false-positive stopped transactions and human labor. Furthermore, it can be integrated seamlessly with the current watchlist filtering system by direct connection to the software's database.

Money transactions traffic decisions and all other related information have been considered as the primary sources of data in order to update the watchlist filtering system database. Their proposed ML-Component then uses this data to assess and analyze historical transactions before making a final recommendation. Finally, they applied multiple machine learning algorithms such as SVM, decision tree, and naïve Bayes on the recommended transactions produced by ML-Component and discovered that SVM outperformed the other techniques.

An improved version of the One-Class SVM algorithm is introduced by Tang and Yin [57] to detect criminal actions of customers. Financial transaction data has an excessive number of dimensions, and they claimed that the SVM algorithm is used in their study since it is less sensitive to high-dimensional data. One-Class SVM tries pre-learning before using a nonlinear transformation to convert input data into a high-dimensional space. Their performance results demonstrated that, despite having a low false-positive rate, the suggested method's accuracy might be enhanced by incorporating more valuable features. One-class SVM requires the user to define the percentage of outliers from the data set, and this quantity may vary based on the data set.

Liu and Zhang [58] introduced the use of scan statistics as a malicious activity detection method. Financial institutions can employ their framework to identify the suspicious sequences on the transaction level. The statistical learning techniques were applied by Villalobos et al. [59] in order to meet the AML requirements of a medium-size bank for complying with the recent regulations. Another machine learning model was developed and proposed by Jullum et al. [60] to prioritize which financial transactions should be manually investigated for potential money laundering. Vu [61] presented a money laundering detection model to classify bank transactions using deep learning, random forest classifier, logistic regression, and SVM. Naive Bayes classifier was also applied in the work of Kumar et al. [62] for transaction classification with the help of big data analytic techniques. Zand et al. [63] designed an AML scheme allowing banks to share information with each other while maintaining the confidentiality and integrity of data. Their framework utilized a neural network as the machine learning model for the detection phase.

A clustering technique called CLOPE [92] is proposed by Cao and Do [93] to identify money laundering cases. CLOPE can only work with nominal variables, and its main idea stems from a realistic situation of actual life data. Paula et al. [94] employed deep learning unsupervised models to detect money laundering cases in Brazilian exports. Also, they used AutoEncoder [95] to detect suspicious situations based on the patterns in data. In another paper, the idea of using a core decision tree for AML was introduced by Liu et al. [96]. In fact, the core decision tree is an improved version of K-

TABLE I
COMPARISONS OF PUBLICATIONS ON AML WITH SUPERVISED MACHINE LEARNING (T=TRANSACTIONS, C=CUSTOMERS, CR=CUSTOMER RELATIONSHIPS, TB=TRANSACTIONAL BEHAVIOR)

| Ref. | Technique | Data Type | | | |
|---|---|---|---|---|---|
| | | T | C | CR | TB |
| [64] | SVM, RBF Kernel | | ✓ | | ✓ |
| [65] | SVM | | | | ✓ |
| [66] | Decision Tree | | ✓ | | |
| [67] | Neural network, Genetic algorithm | | | | ✓ |
| [68] | KNN | ✓ | ✓ | | |
| [52] | Neural network, SVM, Outlier detection | ✓ | | | |
| [69] | Naïve bayes | | ✓ | | |
| [58] | Scan statistics | ✓ | | | |
| [70] | SVM | ✓ | ✓ | | |
| [71] | Graph mining, Fuzzy matching | | | | |
| [72] | Decision tree, Random forest | | | | ✓ |
| [73] | Fuzzy logic | ✓ | | | ✓ |
| [74] | Rule-based, Naïve bayes | | ✓ | | ✓ |
| [75] | Neural network, Fuzzy logic | | ✓ | | ✓ |
| [76] | Graph clustering, frequent pattern, Sequence miner, visualization | | ✓ | | ✓ |
| [77] | KNN, Random forest, SVM | | | ✓ | |
| [78] | Decision tree | | ✓ | | |
| [79] | SVM, Random forest, Logistic regression | | ✓ | | |
| [59] | C5.0, CHAID, C-RT, Neural network, SVM | ✓ | | | |
| [80] | Deep learning models | | | ✓ | |
| [81] | Machine learning cooperating layering simulation (McLayS) | | ✓ | | |
| [82] | Dictionary learning, Graph Laplacian classification | ✓ | | | ✓ |
| [83] | Federated learning, Graph learning | ✓ | ✓ | | |
| [60] | XGBoost | ✓ | | | |
| [84] | Relational KNN, Decision tree | | | | ✓ |
| [61] | Deep learning, Random Forest, Logistic Regression, SVM | ✓ | | | |
| [85] | Logistic regression, Gradient boosting trees | | ✓ | | |
| [86] | SVM, Random forest, Decision tree | | ✓ | | |
| [62] | Naïve bayes | ✓ | | | |
| [87] | Natural language processing, Sentiment analysis | | | ✓ | ✓ |
| [63] | Fast Tree binary classification | ✓ | | | |
| [88] | Random forest | ✓ | ✓ | | |
| [56] | SVM, Decision tree, Naïve bayes | ✓ | | | |
| [89] | SVM, Random Forest, Decision Tree, Naive Bayes, Linear Regression | ✓ | | | |
| [90] | SVM, Random Forest, Artificial Neural Network, Linear Regression | ✓ | ✓ | | |
| [91] | Random Forest, Decision Tree, Linear Regression, SVM | ✓ | | | |

means, and BIRCH [97] which is a hierarchical clustering over particularly large datasets.

One of the recent publications by Tundis et al.[89] proposed a model for enhancing the detection of malicious transactions related to money laundering. Their model was constructed based on defining a group of transaction-related features such as the time, the amount of money, number of transactions, type of operations, and level of internationalization. To evaluate the goodness of their feature-based machine learning model aiming at detecting suspicious transactions, they utilized five different supervised classifiers. Random forest outperformed the other four models, according to their findings. Changjie et al. [91] firstly used practical component analysis (PCA) and visual analysis to obtain the features of transactions. Then, they applied a number of conventional supervised machine learning models to classify the transactions. In their final step, they compared the performance of these models with shallow neural networks.

Unsupervised machine learning models have a stronger capacity to uncover anomalous patterns in new transactions than supervised models, according to Chen et al. [98]; however, their detection results are not superior to supervised models. They presented a set of unsupervised and deep learning approaches to create an AML and fraud detection system to tackle this limitation. Their system leveraged three deep learning models: autoencoder (AE), variational autoencoder (VAE), and a generative adversarial network. To capture time-related fraud patterns, they first preprocessed the given dataset, breaking down the transaction date attribute into its base components. The Wasserstein Generative Adversarial Network (WGAN) was then used to generate fraud transactions, which were mixed with the base dataset to create a more balanced mixed dataset. Next, the AE and VAE models were trained using these two datasets. They also developed two variants of the AE model (single-loss and multi-loss), as well as a novel technique of computing the anomaly score threshold known as Recall-First Threshold (RFT), which improves the model's performance.

### *B. Fraudulent accounts and non-fraudulent accounts*

Individuals constitute a crucial part of the money laundering process. Each customer has a unique account in the banking system consisting of all their personal information and their money movement history. Some works have been accomplished on finding the individual who has the highest level of risk in suspicious activity. In such models, the customer-related features, for example, the individual salary information, account history, and risk value, are considered as input. Then, the account with the highest possibility of being a money launderer is provided in the output. We will summarize published papers of this group in the following.

Bitmap index-based decision tree is one of the AML solutions provided by Jayasree and Balan [78] for the purpose of risk evaluation on money laundering activities. This framework used an indexing scheme to determine the associated risk, which could enhance adaptability and scalability rates. Rows and columns in bitmap indexing stored the information, and then a decision tree was built for mapping the binary fuzzy form to create rules.

The indexing focused on providing pointers to the rows in a table containing given vital values. They showed that money laundering accounts could effectively be evaluated with bitmaps by using cardinality rows and columns, where the bitmap index applies the arrays and bitwise logical operation for the results. The total number of integers in bitmaps was calculated to achieve the population frequencies for risk assessments. Results were finally employed to construct a decision tree where the customer region of transactions and risk occurrences are detected.

The expanding importance of Know Your Customer (KYC) is highlighted by the rapid expansion in the volume of data

and the increased range of risk sources. The research of Chen [86] used machine learning models to develop a risk detection benchmark that can identify the drivers of risks via KYC data, with the benefit of assisting managers in complying with daily new requirements in banking systems. In addition, Chen [86] proved that banks are the best tier in AML for determining risk levels.

One probable explanation is that certain bank branches are located near illegal or unauthorized businesses, putting them in a position to deal with dangerous consumers. The proposed approach of Chen [86] applied SVM and decision trees for the classification of accounts. Since individuals and company accounts have similar vital characteristics, both of their classification methods used these identical features to classify them as individual customers or companies. This method allows managers and regulators to surveillance those particular branches to boost the overall compliance while reducing their legal exposure.

Wang and Yang [66] determine the risk rank of customer accounts in a commercial bank and create a determination rule of money laundering using the decision tree method. To improve the process of money laundering detection, the active learning model is proposed by Deng et al. [69] via a sequential design. Their approach utilized a combination of stochastic approximation and optimal design methods. It is proposed by Zhou et al. [79] to identify the illegal and benign behaviors of accounts based on the multi-faceted feature, which classifies accounts from three aspects: account viability, transaction sequences, and spatial correlation among accounts.

Machine learning cooperation layering system (McLayS) was designed by Tai and Kan [81] to detect suspicious money laundering accounts. McLays is a two-phase intelligent system that identifies fraudulent accounts in the first phase and then detects the most suspicious ones through the second stage. Pavlo et al. [85] provided a two-layer AML architecture addressing two essential requirements: scalability and resistance to imbalance. The first layer detected high confidence non-illicit customers with the help of a logistic regression algorithm. Then, the second layer employed gradient boosting in order to classify the remaining accounts of customers.

### *C. Money laundering networks vs non-money laundering networks*

Money laundering is not an individual operation; however, it involves a group of people working together. Therefore, it would be very productive if we could find relationships between customers to hinder their activity and detect criminals. Some works attempt to find the money flow between accounts and then select those accounts with the highest risk of being fraudulent. The input of these systems is usually a graph, network, or cluster, which represents how money is replaced between different customers. They provide fraudulent relationships or customers as the output. In this section, we will describe published works that apply customer relationships as the main feature to find money laundering cases.

One of the existing works of this group is the detection system given in the research work of Luo [99] who introduced a method for AML consisting of three key components: clustering, frequent pattern mining, and data visualization. The clustering component only captures irregular money transfers and builds a graph reflecting the money flaw between groups of accounts. The task of the second component is to apply mining algorithms in order to extract frequent patterns inside the clusters. Then, they tried to visualize the patterns and money flows between the suspicious accounts. They claimed that mining habits could help find suspicious transaction patterns in nearby clusters. This system may be practical only when dealing with fraudulent activities that occur among a group of people.

The work of Drezewski et al. [76] had resulted in developing a Social Network Analysis (SNA) based algorithm, which is called Money Laundering Detection System (MLDS). In order to assign roles to the accounts associated with the money movement network, they defined six basic metrics for their proposed MLDS, including closeness, betweenness, page rank, degree, authoritativeness, and hubness. The purpose of these metrics is to determine the relationships between accounts and the strength of their connections depending on their financial actions. A social network graph can be generated and visualized once the roles of the accounts have been specified. They reported that algorithms that assess the relationships between roles are far more scalable and outperform traditional role-finding approaches.

Colladon and Remondi [100] designed a system that integrates SNA into an existing factoring company. Their method employed four peculiar relational graphs with different characteristics to create the network of associated accounts for each transaction. According to their findings, a suspicious profile has a lower centrality to its link graph and frequently operates in a more significant number of transactions. Although this method yields reasonable results, it was developed for a factoring company; thus, it cannot be relied upon to deliver convincing results when dealing with massive data sets of transactions in the banking industry.

Due to the efficiency of social network analysis methods in modeling the relationships and building graphs, they have been applied in this category much more than in other categories. In another published article, Drezewski et al. [101] constructed a social network from bank statements and National Court Register. They attempted to assign roles to each individual to find actual leaders, vulnerabilities inside the network and investigate the connections between them more thoroughly. Using a combination of network analysis and supervised machine learning methods, Savage et al.[77] proposed an approach to detect money laundering cases based on the analysis of group behaviors.

Han et al. [80] presented a distributed framework to apply and visualize various natural language processing technologies based on deep learning to enhance money laundering investigation. In another deep learning based method, a convolutional neural network was developed by Weber et al. [102] as a scalable graph for forensic analysis of financial data to map the structural and behavioral dynamics driving the technical challenge. Their network predicts the suspiciousness of a given target node and identifies other potential bad actors in the

TABLE II
COMPARISONS OF PUBLICATIONS ON AML WITH UNSUPERVISED MACHINE LEARNING (T=TRANSACTIONS, C=CUSTOMERS, CR=CUSTOMER RELATIONSHIPS, TB=TRANSACTIONAL BEHAVIOR)

| Ref. | Technique | Data Type | | | |
|---|---|---|---|---|---|
| | | T | C | CR | TB |
| [103] | Clustering | ✓ | | | ✓ |
| [104] | Minimum spanning tree clustering | ✓ | | ✓ | |
| [105] | K-means | ✓ | | ✓ | |
| [93] | Clustering (CLOPE) | ✓ | | | |
| [54] | K-means, Expectation maximization | ✓ | | | |
| [106] | Clustering, Expectation maximization, SVM, Decision Tree | | | | ✓ |
| [107] | K-means | | | ✓ | ✓ |
| [94] | Deep learning, Linear principal component analysis (PCA) | ✓ | | | |
| [96] | Decision tree, K-means | ✓ | | | |
| [108] | One-class SVM, Isolation forest, Gaussian mixture models | ✓ | ✓ | | |
| [109] | Isolation forest, One-class SVM | ✓ | | ✓ | |
| [110] | triage model, rule-based | | | ✓ | ✓ |
| [98] | Autoencoders, Generative Adversarial Networks | ✓ | | | |

TABLE III
COMPARISONS OF PUBLICATIONS ON AML WITH SEMI-SUPERVISED MACHINE LEARNING (T=TRANSACTIONS, C=CUSTOMERS, CR=CUSTOMER RELATIONSHIPS, TB=TRANSACTIONAL BEHAVIOR)

| Ref. | Technique | Data Type | | | |
|---|---|---|---|---|---|
| | | T | C | CR | TB |
| [111] | K-means, Genetic algorithm, Neural network | | | ✓ | ✓ |
| [101] | SNA, Graph mining, K-means | | ✓ | ✓ | |
| [112] | SNA, Graph mining | | ✓ | ✓ | |
| [113] | K-means, ANN | ✓ | ✓ | | ✓ |
| [114] | Fuzzy logic, Naïve bayes | ✓ | ✓ | | |
| [102] | CNN | | | ✓ | |

transaction network through direct or indirect connections to nodes known to be suspicious.

### D. Suspicious behavior vs non-suspicious behavior

Money laundering occurs through the movement of money to different places (or accounts) using different types of transactions. Tracing the spots where cash is deposited or withdrawal gives us an insight into the transactional behavior of customers. Hence, it is helpful to monitor the transactions, identify unusual movements, and make distinctions between normal and anomalous behavior in money transactions. Several researchers have been published in recent years, focusing on customers' behavior in accomplishing banking transactions. This section will be focused on summarizing published AML works related to this taxonomy.

One of the earlier works of this group was introduced by Yang et al. [67]. Using neural network algorithms, they provided an AML service system for a Union Bank to detect money laundering on online payments. The logical framework of their proposed system includes five sequential layers: database layer, basic data resource base layer, data analysis layer, application service layer, and the interface layer. The database layer plays the role of a warehouse where all the information of transactions is accumulated. The primary data resource layer contains a knowledge base, a case database, and other helpful information that allows money laundering cases to be discovered. The data is then cleaned at the analysis layer.

The results are delivered to multiple agents for further analysis, including a neural network, an expert system, and a data mining agent. The detection process occurs in the application service layer, where essential data from new transactions are retrieved and displayed to customers to help them make the best possible decision. Union Bank will be notified if money laundering is discovered. The standard interface layer is just a way for all financial institutions to present transactions to the Union Bank.

Borrajo et al. [84] proposed a mechanism of generating diverse, realistic, and new unobserved behaviors to proactively find and mitigate potential unseen adversarial actions. They explained that customers' behaviors in the environment, which are partially observable by the bank, are always motivated by a number of factors. These behaviors rely on some hidden human goals and states they encountered when trying to obtain these goals. Then, they assumed that goals, states, and actions could be converted in the form of high-level representation, which is in line with the requirements to file rationale for SARs. Indeed, SARs provide information linked to the descriptions of actions taken by suspicious persons and the corresponding states.

Although all of this knowledge might be contained in the attribute-value representation used by most artificial intelligence models for AML, the size of this representation should be normally constrained. Their research aimed to map the AML problem to a relational classification task relying on the KNN algorithm. The input of their proposed system was a trace of human behavior related to the observable actions by the bank and associated observable components of the states. In the output, their system generated an answer on whether the considered trace is part of a money laundering activity.

Outlier detection has a wide range of applications and is one the most popular methods in the field of AML. Jun [65] employed an outlier detection technique based on distance definition incorporated with financial transaction data features which compare each transaction against a peer group to determine whether the transaction behavior is suspicious. An approximation algorithm was also integrated with their model, which has the goal of optimizing the deviation-related calculation from the tested data point to the reference dataset. Multi-agent based simulation technique was applied in the work of Lopez-Rajas et al. [72] to generate synthetic data and then find anomalies in the mobile money domain using the decision tree method.

An unsupervised model was designed by Chitra and Subashini [106] with the help of data mining techniques. The objective of their work was to analyze the applications of data mining techniques in fraud prevention and detection, customer retention, marketing, and risk management. In the next section, we will describe those AML methods that use a combination of data types, based on our taxonomy, as their input data.

### E. Hybrid Models

According to our analysis, the combination of two or three data types has been employed in some of the papers. In other

words, these articles utilize the features of several data types in parallel as their input to detect a suspicious account or transaction in the output. This category is referred to as hybrid models to better categorize the published articles. This section gives a brief overview of each paper in the hybrid models taxonomy.

Relying on customer accounts information and their transactional behaviors as input data, Drezewski et al. [76] designed a money laundering detection system to facilitate the investigation of the analyst. Their proposed system consists of two main components: data importer and analyzing algorithms. First, the data importer component was employed by the clustering algorithms as a basis for money transfer analysis. Then, the analyzing algorithms such as transaction mining and frequent pattern mining were used to extract suspicious patterns in the input data. Tang and Yin [64] also used the same input data type to identify unusual customer behaviors using the SVM classification algorithm. Moreover, Heidarinia et al. [75] developed an adaptive neuro-fuzzy inference system based on user features and their financial actions in order to determine the riskiness of the user behavior.

Due to the fact that money laundering is a group activity, one of the most valuable data is the investigation of customer relationships. To discover money laundering profiles in the bank, some published works use customer account data and relationship information. However, once questionable clients and transactions have been identified, it might be challenging to determine their relationships. Using social network analysis techniques, Shaikh et al. [112] tackled this specific difficulty with a model capable of spotting relationships between unlawful transactions and suspicious customers.

The simultaneous use of customer relationships and their transactional behaviors has been proposed in some of the AML papers. Alexandre et al. [107] developed a system that is capable of detecting money laundering cases automatically and generating signals for suspicious transactions. They used the definition of an agent as a person who will behave as autonomously as feasible in the decision-making process to flag fraudulent transactions. In another research, the use of disorganized detection for AML with the help of sentiment analysis techniques was proposed by Thi et al. [87]. They evaluated data concerning suspicious behaviors, clients' relationships, and the awareness for the customers' retrieval from the financial sector in social media platforms. Le Khac et al. [111] employed data-mining based approaches to analyze transactional data. They provided AML solution via the use of multi techniques such as clustering, neural network, genetics algorithm, and heuristic models.

In a recently published study by Eddin et al. [110], the triage model was offered as a machine learning component that operates on generated alerts of rule-based methods to minimize false positives or prioritize true positives. Their triage approach works hand in hand with the rule-based system, learning to accurately forecast the risk of an alert. In addition, they used graph-based features to characterize inter-entity relationships (customer relationships) as well as entity-centric (customer transactional behaviors) designed features.

Compared to other hybrid models, the use of raw transaction and customer profiles data have been suggested more in the reviewed papers. In order to obtain risky transactions, probabilistic models were deployed in the work of Liu et al. [68]. They tried to seek out malicious sequences by analyzing transactions history and customer-related information. Cross validation [115] is a technique that tries to find optimal parameters based on the classification accuracy via grid search. Keyan et al. [70] attempted to select better parameters for SVM with the help of cross validation and classify transactions. Federated graph-learning is a ML-based method implemented by Suzumura et al. [83] for AML. The goal of their federated model was to share insights across multiple financial institutions without sharing any raw data from each financial institution.

In addition, Ketenci et al. [88] suggested to consider time-frequency features to simplify the feature selection step and improve the data science method of AML. Analyzing financial information without the use of ground truth is the other methodology for money laundering detection proposed by Camino et al. [108]. Their method generated a user suspicion score which helped the human experts in order to validate customers and their transactions through an ensemble of anomaly detection algorithms. SARDBN (Suspicious Activity Reporting using Dynamic Bayesian Network) was a model introduced by Raza and Haider [114] which utilized clustering and dynamic Bayesian network (DBN) to detect anomalies in the sequence of transactions. Their proposed SARDBN found patterns in monthly transactions of customers and calculated the anomaly index called AIRE (Anomaly Index using Rank and Entropy). AIRE measured the degree of an anomaly in a transaction and compared it against a pre-defined threshold to label the transaction as legitimate or suspicious. The recent work of Omri [90] employed SVM, random forest, artificial neural network, and linear regression to detect anomalous cases of money movements in a supervised manner.

The transaction-related information includes a wide range of meaningful signs to find fraudulent money transfers. It is proposed by some works to consider transactional behavior of customers as a supplement to transactions details for identifying suspicious movements of money. Chen et al. [73] used transaction data and suspicious behavior of customers to implement a fuzzy computing system for combating money laundering. Dictionary learning methods were applied in the work of Baltoiu et al. [82] to exploit the structural information in graphs, improve the classification and detect anomalous cases in transactions. The idea of empirical mode decomposition was introduced by Zhou and Tianging [103] for finding fluctuation variances in financial time series and using them as a basis for detecting suspicious transaction behavior.

Another group of hybrid models suggested to use the characteristics of financial transactions while monitoring the relationship of associated customers of those transactions. Dissimilarity was applied by Wang et al. [104] as a metric for measuring the difference degree of an outlier in transactions. Their measurement results were then utilized to design a money laundering detection framework based on the improved minimum spanning tree algorithm [116]. Shokry et al. [109] in another published work, introduced an unsupervised machine learning model which is able to detect hidden patterns, groups,

and transactions of money laundering promptly to counter terrorism finance. The use of the clustering technique was suggested by Larik and Haider [105] in order to establish normal behaviors of customers. Also, they applied statistical techniques to evaluate the deviation of a particular transaction from the corresponding group behavior.

A few number of recently published papers used more than two data types as input of their model. Transaction details, customer profiles information, and transactional behaviors of customers are three data types that were applied in some of the reviewed papers of this survey. Using these data types, a Bayesian model was designed by Nida S. Khan et al. [74] which assigns a money laundering score to each customer as an indicator of customer transactional behavior. Then, if there is a remarkable difference in their behavior, an alert is generated, which should be checked by an expert to verify whether it is money laundering. Le-Khac et al. [113] presented a framework based on the combination of clustering and classification techniques to analyze money laundering patterns in an international investment bank. It was mentioned in their work that customer behaviors are complicated in such a bank because it is influenced by several factors.

## V. Analysis of Existing Work

In the following paragraphs, the summary of our investigation is provided from different perspectives, including the type of dataset and data-related challenges, the type of algorithms used in the literature review, and the performance of the models.

### A. Dataset

Based on the existing literature reviews, it can be observed that most works use real datasets to validate their approach in the test phase. While real data can measure the method's applicability in the real financial environment, it is observed that most of these real datasets only contain benign transactions, and therefore money laundering cases are artificially generated. In fact, only a small number of published works have a real dataset of both benign and suspicious cases [68][66][73][34][52][114][99]. The majority of other approaches used a mixture of real standard transactions dataset along with synthetic money laundering data. Figure 5 illustrates the distribution of existing literature works based on the dataset type. Also, the category-wise distribution of the published articles over the last two decades is illustrated in Fig 4.

The ability of a machine learning approach to analyze and process a vast volume of complex transaction data is directly proportional to the dataset size. According to our literature research, a significant portion of machine learning models were validated on a small dataset of less than 10,000 transaction records. As a result, they demonstrated considerably rapid performance for the entire detection process, despite the fact that this appears to be far from real-life scenarios. Therefore, it is difficult to rely on their performance time, which makes them an improper candidate for the purpose of evaluation and comparison. In fact, these methods need to be re-evaluated on an adequate number of data to assess whether they are capable of scaling to a considerable number of transactions dataset.

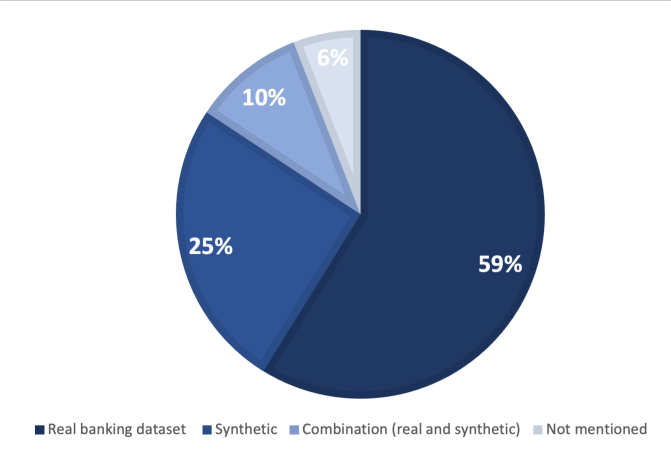



Fig. 3. Distribution of existing works based on their dataset.

Furthermore, financial datasets have been discovered to be largely suffered from missing data since they contain private information that might be hidden by the financial institution or even not provided by the customers. For example, customers may be unwilling to provide personal information such as their home address, phone number, and other personal information to financial institutions. However, most of the works focus solely on discussing the phases of model implementation and do not provide or even indicate how to deal with missing values.

The dilemma of class imbalance of financial datasets is another significant concern in the AML field. As previously stated, the majority of the datasets utilized in the publications were either actual samples from a specific bank or a mix of real and intentionally generated samples. In real-world datasets, the number of legal transactions is remarkably higher than the number of suspicious transactions, resulting in a class imbalance and a significant false-positive rate. Hence, it is crucial to have some reasonable solutions for questions such as how to deal with class imbalance, how synthetic datasets are generated, and which generating method leads to better results. It is worth noting that, despite the importance of class imbalance and synthetic data generation, published studies do not go into great length about these.

Most of AML approaches rely on the same set of features. These features include account transaction frequency within a particular time window (i.e., daily, weekly or monthly), amount withdrawn, amount received, and any other transaction-related information. In addition, some other published studies designed their methods based on the customer-related data and attributes such as individual salary information, sender's or receiver's account history, and risk score associated with the customer account. These data are referred to as input data, and they are used to construct our proposed taxonomy in this survey.

TABLE IV
COMPARISON ON DATASET AND RESULTS OF PUBLISHED WORKS

| Ref. | Real Dataset | Synthetic Dataset | Training Size | Test Size | Accuracy | False Positive/True Positive |
|---|---|---|---|---|---|---|
| Data type: Transactions | | | | | | |
| [52] | ✓ | ✓ | 200(clean),70(dirty) | 50(clean),40(dirty) | 0.87 | 0.76/1 |
| [58] | ✓ | - | 120,986(clean), 1,797(dirty) | - | - | 0.051/- |
| [59] | ✓ | ✗ | - | - | 0.996 | -/- |
| [60] | ✓ | ✗ | 13,782 | 2,410 | 0.907 | -/0.95 |
| [62] | ✗ | ✓ | 7,000 | 3,000 | 0.8125 | -/- |
| [56] | ✗ | ✓ | - | 1500 | 0.848 | 0.118/0.815 |
| [93] | ✓ | ✓ | - | 12,350(clean),25(dirty) | - | -/- |
| [54] | ✓ | ✗ | 2035(clean)31(dirty) | 665(clean)14(dirty) | - | 0.016/0.93 |
| [94] | ✓ | ✗ | - | 819,990 | - | -/- |
| [96] | - | - | - | - | - | -/- |
| [89] | ✗ | ✓ | - | - | 0.9544 | 0.067/0.972 |
| [98] | ✓ | ✓ | 3799(clean), 106(dirty) | 950(clean), 27(dirty) | 0.93 | 0.07/- |
| [91] | ✗ | ✓ | 1394 | - | 0.992 | -/- |
| Data type: Customers | | | | | | |
| [66] | ✓ | - | 160,000 (12%dirty) | - | - | -/- |
| [78] | ✓ | ✗ | - | 350 | - | 0.6238/0.9 |
| [79] | ✓ | ✗ | 381,523(clean),114,891(dirty) | - | 0.942 | 0.0097/- |
| [81] | ✓ | ✗ | 4,389 | 4,051 | 0.8704 | -/- |
| [85] | ✓ | ✗ | 230,000 | 100,000 | - | 0.0066/0.4807 |
| Data type: Transactional Behavior | | | | | | |
| [65] | ✓ | ✓ | - | 80(dirty) | 0.65 | -/- |
| [67] | ✓ | - | - | - | - | -/- |
| [72] | - | ✓ | 486,977(clean), 6,006(dirty) | - | 0.99 | 0.009/0.999 |
| [84] | ✗ | ✓ | - | - | 0.9 | -/- |
| Data type: Customer Relationships | | | | | | |
| [77] | ✓ | ✗ | 530,790 | 227,481 | - | 0.4/0.94 |
| [80] | ✓ | ✓ | 12,467 | - | 0.88 | -/- |
| [102] | ✗ | ✓ | - | - | - | -/- |
| Data type: Hybrid | | | | | | |
| [64] | ✓ | ✓ | 32000(clean),30(dirty) | - | 0.678 | 0.034/- |
| [68] | ✓ | - | 120,986(clean),1940(dirty) | - | - | 0.558/0.954 |
| [70] | ✓ | ✓ | 32,000(clean),1930(dirty) | - | 0.63 | 0.06/0.79 |
| [73] | ✓ | ✗ | - | 710 | - | -/0.1 |
| [75] | ✗ | ✓ | 1500 | 500 | 0.96 | -/- |
| [76] | ✗ | ✓ | - | - | - | -/- |
| [82] | ✗ | ✓ | 4800(clean),480(dirty) | 1200(clean),120(dirty) | 0.9977 | -/- |
| [83] | ✗ | ✓ | - | - | 0.995 | -/- |
| [87] | ✗ | ✓ | 560 | 140 | 0.851230 | -/- |
| [88] | ✓ | ✗ | 4,893(clean)1787(dirty) | 4,236(clean)995(dirty) | - | 0.1142/0.7415 |
| [103] | ✓ | ✗ | - | 1042 | - | -/- |
| [104] | ✓ | ✓ | - | 65,001(clean)60(dirty) | - | -/0.72 |
| [105] | ✓ | ✗ | - | - | - | -/- |
| [107] | ✓ | ✗ | - | - | - | -/- |
| [108] | ✓ | ✗ | - | - | - | -/- |
| [109] | ✓ | ✗ | - | 118,250 | - | -/- |
| [111] | ✓ | ✗ | 3000(clean)216(dirty) | - | - | -/1 |
| [101] | ✓ | ✗ | - | - | - | -/- |
| [112] | ✓ | ✗ | - | - | - | -/- |
| [113] | ✓ | ✗ | 200K | 1.8M(clean)10(dirty) | - | -/1 |
| [114] | ✓ | ✗ | - | - | - | -/- |
| [110] | ✓ | ✗ | 300,000 | 200,000 | - | 0.2/- |
| [90] | ✗ | ✓ | - | - | 0.99 | -/0.21 |

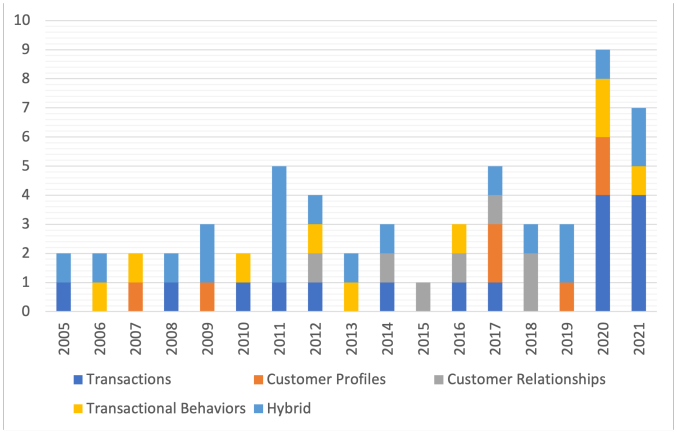



Fig. 4. Category-wise distribution of AML papers in each year.

### B. Performance

According to the literature review, the most time-consuming part of the AML techniques is the learning process which is ineffective when dealing with a large volume of data. The main reason is the training approach of AI models. They re-train the entire model whenever they receive new data. One possible solution to address this issue is the use of reinforcement learning [117] which maintains a continuous learning process on live input data. Reinforcement learning allows an agent to learn through trial and error in an interactive environment based on the feedback from its own actions and experiences.

The proposed method of Lv et al. [52] can be considered as another effective solution to tackle this problem. The main advantage of their model is the ability to adjust to transaction data that hasn't been seen before. When new unseen transactions data are introduced, their method dynamically fine-tunes the ML models' clustering radius, allowing the system to run smoothly without the need for training from initial.

One of the most commonly used performance metrics in existing work is accuracy, as can be seen in Table IV. However, accuracy is not a suitable metric for evaluating the performance of AML models, because AML datasets are extremely imbalanced. For example, the ratio of money laundering transactions to legitimate transactions in the PAYSIM synthetic dataset [118] on Kaggle is approximately 1000:1. In real world data, this ratio is even much smaller [18].

In a large number of papers, the confusion matrix was used to compare the proposed methods to their ground truth. Also, the detection rate, along with the true positive and false-positive rates, was commonly measured to show the AML approach's improvement. In some articles, the AUROC graph was provided in order to intuitively demonstrate the performance of the transaction classifier. However, the problem of an imbalanced dataset can potentially mislead the performance to a high degree of accuracy, as the majority of the output will readily fall into the benign class of transactions.

It is observed that AUROC can provide a trade-off between specificity and sensitivity. As a result, the optimal AML method can be easily identified: one with a curve that is closer to one and further away from the threshold line. Although this may be true, it is essential to further validate the method because they mostly use synthesized small datasets.

### C. Algorithm

From the work of Tang and Yin [57] in 2005 through the work of Jullum et al. [60] in 2020, there have been lots of technological advancements in statistical methods, machine learning to deep learning methods, and other AI technologies in the field of AML. Despite the robustness of these technological improvements, the severity of the suspicious money laundering transactions detection problem has stayed relatively stable over the last two decades. Many articles have proposed the application of these methods for detecting doubtful cases of money movement. They have reported double-digit improvement in their performance; however, there are little shreds of evidence of practical use of these methods by relevant entities such as financial institutions and actual performance in terms of false-positive alert reduction or operational cost reduction [119].

The majority of the articles surveyed prefer to use approaches that rely on previously detected fraudulent transactions. Furthermore, anomaly detection techniques are considered as a common solution for the AML field. Although they have been suggested and used in the literature review, it is important to remember that they merely detect transactions that are deviated from the norm. These variations could be the result of suspicious transactions, but they could also be the result of other factors that should be analyzed by a professional.

It is further observed that decision tree and its variants such as random forest [72], Adaboost, or inspired bitmap-based [78] are the most commonly used solutions in rule-based approaches. SVM and neural networks are the two most extensively applied algorithms for the purpose of classification in the AML context. The alternation of kernel algorithm in SVM (such as RBF [52]) and the depth of layers in neural network configuration can enhance the performance of the detection process. In terms of clustering, K-means has been employed frequently due to its purity and simplicity.

In fact, K-means' contribution in the existing literature review is to compare with other similar techniques or to shorten the learning phase for classification in semi-supervised methods. Instead of using the traditional Euclidean or Manhattan distance measurement techniques, most of the works used a variety of distance measurement methodologies to improve the K-means performance. It's worth noting that some other studies employed approaches like Expectation Maximization, which focus on assessing the likelihood or matching probability. These methods are more likely to produce reliable results, and they can be used alone rather than in conjunction with other detection techniques.

It is also observed that graph neural network is one of the most widely used methods for detecting suspicious transactions by building a graph of money movements. Besides the potential of graph neural networks, there are several challenges [120] that make practical implementation more complicated, such as complexities caused by high-speed transaction systems, real-time systems, multi-channel updates, data size, data

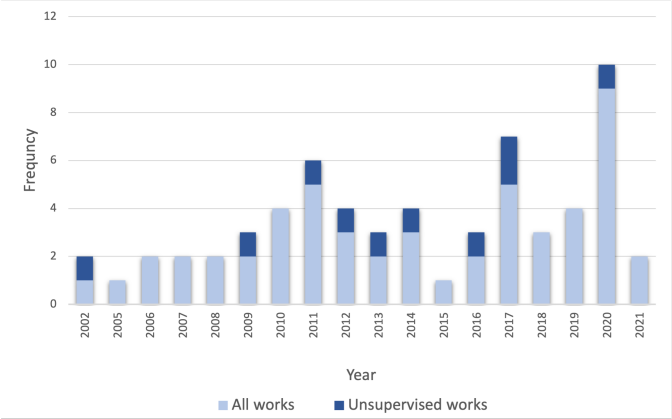



Fig. 5. Distribution of unsupervised methods compared to other available approaches.

speed, data variety, and multiple business applications. As a result, to increase the performance of future graph-based solutions, it is critical to examine the necessary architecture, tools, and considerations.

Based on the findings from literature reviews, it is obvious that the number of supervised approaches is significantly much more than unsupervised methods. Figure 5 shows that supervised machine learning models have been the subject of the majority of recent publications. Unsupervised algorithms appear to be more fit for AML because they can work without labeled data; however, they aren't considered sufficiently in the present literature. In addition, it is hard to find published AML datasets because of the confidential nature of AML data. Hence, the potential of the unsupervised method to perform better in money laundering detection should not be underestimated.

To sum up this section, it is critical to guarantee that AI algorithms, particularly machine learning models, can be used successfully on real-world and large-scale financial datasets. Further research is needed to examine more efficient methods for designing AML solutions. Hence, the opportunity to explore a cutting-edge AI solution that meets the aforementioned requirements is still open.

## VI. Commercial Products

There are many commercial projects available. Lexis-Nexis, Logica ISL, SAS, Oracle AML, NICE Actimize, and Logica ISL are just a few names in this space.

The suspicious activity monitor feature in NICE Actimize has an elaborate library of different models for different use cases and applications such as securities, insurance industries, and banking. This platform also offers extensive analytics and data visualization, and reporting.

SAS supports risk scoring and rule-based decision making. This solution also supports behavioral activities for later assessment with peer-based analytics to enhance decision accuracy.

Traditionally, commercial solutions adopted a rule-based approach to identify potentially suspicious transactions based on a pre-defined set of rules and thresholds. Alerts are triggered when the thresholds are met. More recently, AI has been used in AML solutions. Search space, which was later acquired by Actimize, was an AML solution without any fixed rule. Rather, it relied on support vector machines with probabilistic thresholds to identify anomaly transactions [121]. NICE Actimize claims a 73% money laundering detection rate with a ratio of 1 to 15 false positives[122]. Similarly, Logica ISL uses AI to reduce the false positive rate by over 50% [123].

## VII. Conclusion and Future Research Directions

Financial institutions are considered as the front line to combat money laundering activities. Having effective AML solutions, practices and techniques empower banks to mitigate and reduce the risk of money laundering. Anti-money laundering solutions allow the reduction of the manual steps in the client screening process. The ever-increasing volume of financial transactions and client interactions have made AML compliance highly challenging. Recently, machine learning approaches have proven to provide promising results in detecting and differentiating suspicious cases. This survey provides an overview of machine learning enabled AML solutions and presented them through a new taxonomy model.

The primary objective of AML works is to identify the most effectual approaches to recognize suspicious activities with the highest accuracy and performance measurements possible. In order to achieve these goals, it is critical to execute proper data preparation, data transformation, feature selection, and data analytics. In this survey, we explore all published research work to conduct an in-depth evaluation in terms of ML-based money laundering detection systems. To the best of our knowledge, we have covered all recently published papers.

Based on our findings, class imbalance and the substantial volume of transaction data are common challenges found in most research works, yet only a single paper [124] has recognized or partially addressed this issue. Moreover, some papers [67][111][96][114] suggest to apply semi-supervised techniques as a more suitable approach to handle a significantly large volume of data.

Due to the nature of financial data and privacy policies, most works use synthetic transactions data entirely or as a supplement to real data to train and test their machine learning model. Although unsupervised machine learning techniques may work better when there is insufficient data, the majority of publications have focused on supervised methods. Therefore, unsupervised techniques could be a potential future direction to address the issues around inadequate actual banking data.

Banks produce an immense volume of financial records that range from simple point to point to very complex forms of transactions. In most papers, the proposed AML solutions are required to perform a complete retrain based on all the data, causing repeated delays after the introduction of new records. Hence, the learning process time is another major challenge in the context of money laundering detection. Reinforcement learning techniques are a promising solution capable of extending the training process to cover the newly introduced data.

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